Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 21323

F-423

B.C.A. (V Semester) Examination, 2025-26

(New Course)

Paper - BCA-503-N

Introduction to Internet Programming

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry **equal marks**. Symbols used have their usual meanings.

1. (a) What are the primary components of a Java application? Explain the role of each component in a simple program.

   (b) Explain the process of compiling and running a Java program. What is the importance of the Java byte code in ensuring platform independence?

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2. (a) What are Java primitive and reference data types? Explain each category with suitable examples.

   (b) What are variable identifier naming rules and conventions in Java? Write a Java program to demonstrate the use of constants and variables.

3. (a) What is the difference between object references and primitive variables? How are they created and used?

   (b) Write a Java program to check whether a number entered by the user is even or odd using conditional constructs.

4. (a) What is a constructor? Explain its syntax, types, and purpose with a suitable example.

   (b) Explain the concepts of inheritance and abstract classes in Java. How is multiple inheritance implemented?

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5. (a) Differentiate between method overloading and method overriding with suitable Java examples.

(b) What is encapsulation in object-oriented programming? Explain typecasting and promotion of expressions with examples.

6. (a) Explain different looping constructs in Java (for, while, do-while). What is a for-each loop, and when is it preferred?

(b) Define Graphical User Interface (GUI). Explain the Java A WT class hierarchy and describe how to create a simple GUI using Frame and Button components.

7. (a) What is the difference between client-side scripting and server-side scripting? Discuss their roles in web application development

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(b) What is HTTP? Explain the process of reading an HTTP request and creating an HTTP response with an example.

8. Write short note on any **four** of the following :

(a) Java Virtual Machine (JVM)

(b) Application Programming Interface (API)

(c) Active X Controls

(d) Active Server Pages (ASP)

(e) The import statement in Java

(f) String class and its memory representation

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